Assessment Tool-3

CO_2

Debugging and Optimization

Name:- M. Pranay

Reg No:- 19252382

Course Code:- ITA0612

Course Name:- Machine Learning.

1) Perceptron Model Fails to Converge.

A Perceptron is a supervised learning algorithm used for classification.

Possible causes:-

1. Incorrect learning rate.
2. Improper weight initialization.
3. Errors in bias handling.
4. Incorrect weight update rules.

Debugging and Optimization:-

→ check the weight and bias update formulas.
→ select a suitable learning rate.
→ verify input data and labels.
→ monitor errors during training.

Conclusion:- Correct implementation and Proper learning rate selection help the Perceptron converge faster.

2) multilayer Perceptron shows slow convergence.

A multilayer Perceptron (MLP) uses multiple layers and backPropagation to learn Patterns.

Possible causes:-

1. vanishing gradients.
2. Improper activation functions.
3. Poor weight initialization.
4. Incorrect learning rate.

Debugging and optimization:-

→ monitor training and validation loss.
→ use ReLU activation where appropriate.
→ Normalize input data.
→ use the Adam optimizer.
→ Adjust the learning rate.

Conclusion:- Proper Initialization, normalization and optimization improve convergence and Performance.

5) Genetic Algorithm Produces Poor solutions.

A Genetic Algorith (GA) finds solutions using selection, crossover and mutation.

Possible causes:-

1. Incorrect chromosome encoding.
2. Poor fitness function.
3. Premature convergence.
4. Lack of Population diversity.

Debugging and optimization:-

→ verify chromosome encoding and fitness values.
→ Adjust mutation rate.
→ Maintain Population diversity.
→ Use elitism to Preserve the best solutions.
→ Repeat runs to compare results.

Conclusion:- Proper fitness evaluation and mutation improve solution quality and convergence.

4) Deep neural Network suffers from overfitting.

Deep neural networks learn complex Patterns, while genetic Programming evolves solution to Problems.

Possible causes:-

1. Excessive model complexity.
2. Too many Parameters.
3. Insufficient training data.
4. High computational cost.

Debugging and Optimization:-

→ Compare training and validation Performance.
→ Apply regularization and dropout.
→ Use Pruning to remove unnecessary structures.
→ Increase training data when Possible.
→ Use efficient evaluation methods.

Conclusion:- Regularization, dropout, and Pruning reduce overfitting and computational cost.

5) Decision Tree Has Poor Test Accuracy :-

A decision tree Predicts outcomes using feature-based rules. High training accuracy but low test accuracy indicates overfitting.

Possible Causes:-

1. Excessive tree depth.
2. Noisy training data
3. Irrelevant features.
4. Overfitting to training examples.

Debugging and Optimization:-

→ Compare training and test accuracy.
→ Use cross-validation
→ Apply Pruning to remove unnecessary branches.
→ Limit tree depth.
→ Perform feature selection.

Conclusion:- Pruning and cross-validation reduce overfitting and improve Performance on unseen data.